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Data over light

Intel's latest technology could increase the average available bandwidth by a bit – quite a bit!

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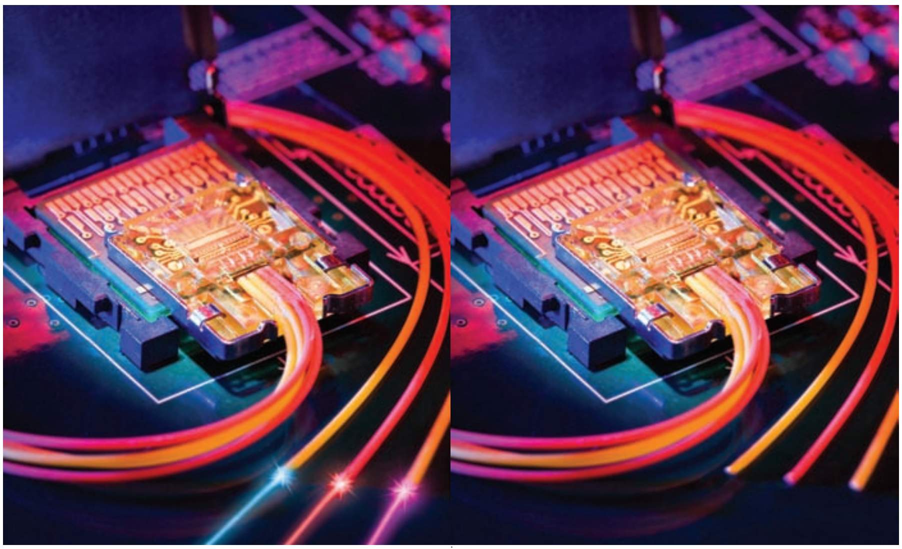
Since the inception every computer user and developer has yearned for more bandwidth either for communication via the internet or to up their own peripheral speeds. Remember that joyous cry you let out while migrating from USB 1.0 to USB 2.0 or if you are on the bleeding edge then from USB 2.0 to USB 3.0. Well Intel claims that they have done better with their aptly named 'Light Peak' technology. We have been using optical fiber for external communication for a long time and now Intel has come up with a technological breakthrough to do the same with peripheral communication. It should be noted that Light Peak and USB 3.0 are complimentary technologies and Light Peak is not designed to kill USB 3.0, at least not during the early stages, though it may be a logical succession as Intel suggested.

Intel describes Light Peak as the codename for an innovative high-speed optical cable technology that is designed to be the interface medium for your electronic devices.

First demonstrated at the Intel Developer Forum in 2009 as a fully functional prototype. To show off their technology, they ran two full HD (or 1080p) video streams, LAN and other storage devices over a single cable that was astonishingly long at 30 metres (demonstrating lack of any signal drop over a significant distance). They clam the same over distances as large as 100 metres for a 125 micron wide cable) and used USB end connectors. In the latest demonstration prototype, the laptop was fitted with a 12-mm square chip as the optical module that performs the optical to electrical encoding. As for the display, there is an external converter box to perform the same function. Light Peak cables are also durable and not prone to interference. According to Intel, "You can tie a knot in it and it'll still work". By May 2010, they demonstrated the same in a laptop signifying that physical dimensions are already small enough for the mobile computation platform. This is when Light Peak became the darling of the tech community.

At its very basic state, Light Peak comprises two optical fibres carrying upstream and downstream traffic separately and dedicatedly. It should be noted that Light Peak isn't protocol dependent, but a mere change in the physical transmission medium. This means it can work on the existing protocols or standards say USB or FireWire with massive boost to their transmission speeds. Current prototypes operate at 10 Gbps in each direction. At 10 Gbps, you can transfer a full length Blu-ray movie in less than 30 seconds.

Given the massive transmission speeds, it also gives developers the liberty to implement functionality to run multiple protocols simultaneously. This means you can connect multiple devices. This is an exciting new feature as this can go a long way in minimising keyboard wire clutter and further miniaturisation, Especially in netbooks and ultra-mobile devices where space is at a premium. Imagine only a single cable except the power plug coming out of your laptop connecting all your peripherals from video, audio, printers to storage solutions. There are design concepts where a tablet performs differently based on the dock style it is connected. With Light Peak that concept might be a reality sooner than you expect since it needs only a single connect. Popular and useful features such as hot plugging and quality of service are also implemented.



Light Peak Close up with Laser switch on and off.